European Journal of Orthopaedic Surgery & Traumatology (2025) 35:395
https://doi.org/10.1007/s00590-025-04510-2

RESEARCH

Quadriceps versus hamstrings autograft in anterior cruciate ligament reconstruction: a randomized control trial

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Received: 1 May 2025 / Accepted: 1 September 2025 / Published online: 13 September 2025

Abstract

Purpose The use of quadriceps tendon (QT) autograft is on the rise owing to its specific attributes, such as predictable size, lower graft failure rate, and less donor site morbidity. The hypothesis for the current study is that ACL reconstruction using QT autograft results in better clinical outcomes than hamstrings tendon (HT).

Methods Patients underwent a single-bundle, all-inside ACL reconstruction after being randomized into QT and HT groups. At 12 months, objective outcomes (Lachman test, anterior drawer test, pivot shift test, and side-to-side anterior translation measurement with KT-1000) and subjective outcomes (Lysholm and KOOS scores) were evaluated, along with donor site morbidity and graft failure.

Results Sixty-nine patients (QT: 33, HT: 36) were included in the final analysis. At 12 month follow-up, a negative Lachman test was found in 72.7% of cases in the QT group and 41.6% in the HT group ($p = 0.04$). Positive pivot shift was more frequent in the HT group (13.9% vs. 0%, $p = 0.27$), while KT-1000 showed greater anterior translation in the HT group than QT (3.77 ± 2.01 vs. 2.73 ± 1.64 mm, $p = 0.23$). Lysholm (92.4 vs. 88.2, $p = 0.008$) and KOOS (94.1 vs. 89.6, $p = 0.01$) scores were higher in the QT group. The only case of graft failure occurred in the HT group, which also had more sensory hypoesthesia (30.5%), while one patient in the QT group developed stiffness requiring arthrolysis.

Conclusion Quadriceps tendon autograft in ACL reconstruction offers superior translational stability and reduced donor site morbidity, with comparable PROMs and a trend toward improved rotational stability.

CTRI registration number CTRI/2023/07/055894.

Evidence Level Level 1, Randomized Controlled Study.

Keywords ACL · ACL reconstruction · Quadriceps tendon autograft · Hamstrings tendon autograft

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Introduction

The ideal graft for anterior cruciate ligament (ACL) reconstruction should have similar biomechanical properties as native ligament, be easily harvested, have the least harvest site morbidity, be secured predictably, and get well incorporated with the bone [1]. The ideal graft is yet to be found, with advantages and disadvantages with each graft. Bone patellar tendon bone (BPTB) graft has historically been considered the gold standard for ACL reconstruction due to its excellent clinical results and patient satisfaction in long-term follow-up [2].

The advantages of hamstrings tendon autograft are no risk of fracture of the patella or tibial tuberosity, avulsion of the patellar tendon, or pain with kneeling, thus minimizing donor site morbidity [1]. Disadvantages of hamstrings autograft include reduced knee flexion strength, saphenous nerve palsy, inferior fixation strength, delayed bone tendon junction healing, and graft elongation [2].

The QT autograft, first described by Marshall et al. in 1979 [3], has emerged as a reliable option for ACL reconstruction. It provides sufficient graft dimensions without violating the suprapatellar pouch [4], contains more collagen fibrils, and demonstrates superior biomechanical strength compared with patellar and hamstring grafts [5]. Clinically, it is associated with lower donor site morbidity than BPTB autografts [6]. However, potential disadvantages include the risk of patellar fracture with bone plug harvest and higher revision rates reported in registry data [7].

The QT remains one of the least studied autografts, although initial results have been promising. The present study aims to provide high-level evidence comparing the clinical outcomes of QT and HT autografts in ACL reconstruction. We hypothesized that ACL reconstruction using a QT autograft would yield superior objective and subjective outcomes compared with HT autografts.

Materials and methods

Study design

A prospective randomized control study was conducted at the Sports Injury Division, Department of Trauma Surgery (Orthopaedics), XXBLINDEDXX. It was a single-center study, and ethical clearance was granted by the Institutional Ethics Committee (XXBLINDEDXX/IEC/23/240). The study has been carried out in accordance with the CONSORT 2010 (Consolidated Standards of Reporting Trials) guidelines and registered under Clinical Trials Registry-India (CTRI/2023/07/055894).

Study population

The study was conducted at a tertiary-level center from July 2023 to October 2024. Patients aged 18–45 years with magnetic resonance imaging (MRI) proven complete tear of the ACL with no other concomitant ligament injury in the ipsilateral knee and no articular cartilage defect were included in the study. Cases where partial meniscectomy or meniscus repair was performed were included. Revision ACL surgery cases, patients with a prior history of fracture around the knee, and patients who refused to take part in the study were excluded. Written and informed consent was taken from all the patients enrolled in the study.

Randomization

After due consideration of inclusion and exclusion criteria, patients were randomized into the QT group and HT group. A simple randomization technique using computer-generated random numbers (Research Randomizer, https://www.randomizer.org/) was done by the senior author (BS), and graft options were revealed inside the operating room immediately before surgery. The data analyst was blinded to the type of graft used for anterior cruciate ligament reconstruction (ACLR) by coding the graft allocation groups and ensuring the knee region remained concealed during patient interactions. Blinding of patients and clinical outcome assessors was not feasible, as QT harvesting required a separate incision. The study procedure is shown in the CONSORT flow diagram in Fig. 1.

Surgical technique

A minimally invasive technique was used to harvest a partial-thickness quadriceps tendon autograft without a bone block [8]. The graft was obtained from the central tendon, with a 2 cm distal strip elevated subperiosteally from the patellar surface. The periosteal end was folded and secured with whip-stitch sutures, resulting in an additional gain of approximately 1 cm in graft length. A 2 mm Hi-Fi® Tape (ConMed, USA) was mounted on a Keith needle and used to secure the graft to the adjustable suspensory button loop. Multiple passes were made through the graft and button loop to ensure fixation, after which the suture ends were tied and buried within the graft. The procedure was repeated at the opposite end, and the final construct was sized with an ACL graft sizer (Stryker, USA) and tensioned on a preparation board (Fig. 2). Hamstring autografts were harvested through a 3 cm anteromedial incision, 2–3 cm distal to the tibial tuberosity. The semitendinosus and gracilis tendons were passed through an adjustable loop button (Procinch; Stryker), sutured together, quadrupled, and secured with looped sutures. Graft diameter was measured using an ACL graft sizer (Stryker, USA), and the construct was tensioned on a preparation board (Fig. 3).

In both groups, grafts measured 6.6–7 cm in length and 7.5–10 mm in diameter. All procedures were performed as all-inside, single-bundle arthroscopic ACL reconstructions via standard anterolateral and anteromedial portals. QT autografts consisted solely of soft tissue, while HT autografts were prepared from semitendinosus and gracilis tendons. Suspensory fixation was performed on both femoral and tibial sides using adjustable loop endobuttons (Procinch, Stryker).

Outcome parameters

Patients were evaluated preoperatively and at 6 and 12 months postoperatively using both clinical tests and patient-reported outcome measures (PROMs). Objective assessments included the Lachman, anterior drawer, and

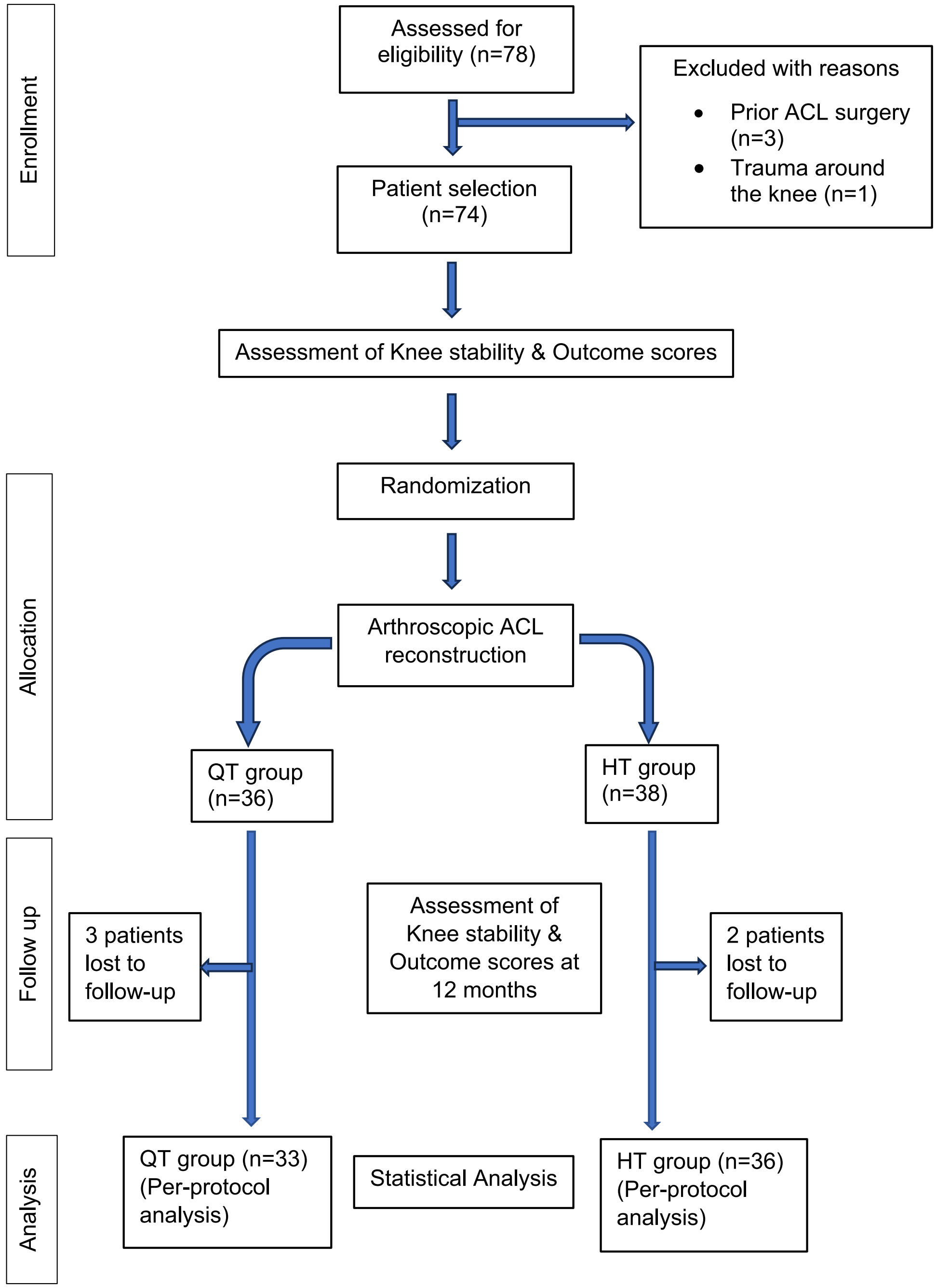

Fig. 1 CONSORT diagram illustrating patient flow

pivot shift tests, as well as side-to-side anterior translation measured with a KT-1000 arthrometer at 12 months. Subjective outcomes were assessed using the VAS, Lysholm, and KOOS scores. The Lachman and anterior drawer tests were graded as: negative/0 (no translation), grade 1 (0–5 mm), grade 2 (6–10 mm), and grade 3 (> 10 mm). The pivot shift test was graded as negative (no subluxation), grade 1 (glide), grade 2 (clunk), and grade 3 (locked subluxation). Graft failure, donor site morbidity, and complications were also recorded. All clinical evaluations were performed by a fellowship-trained consultant orthopaedic surgeon who was not involved in the surgical procedures.

Postoperative rehabilitation

A similar rehabilitation protocol was followed for both groups. Isometric quadriceps strengthening exercises and closed-chain knee range of motion exercises were started on post-op day 1. Weight-bearing as tolerated was started

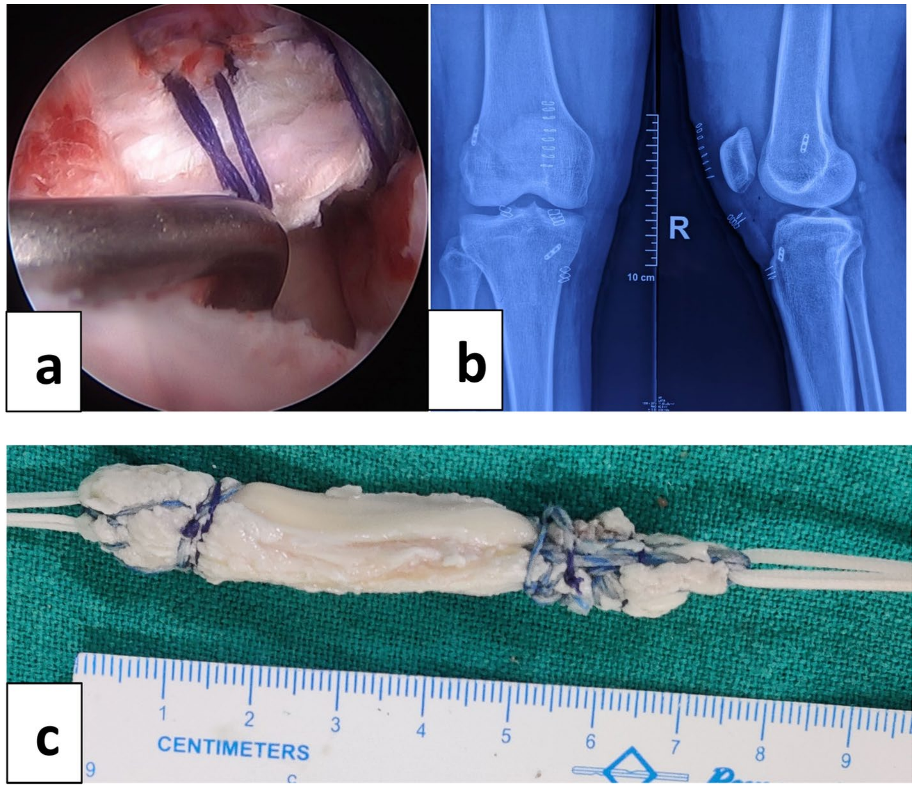

Fig. 2 Illustrating ACL reconstruction with QT autograft; **a** Intraoperative picture, **b** Postoperative X-ray, and **c** Prepared QT autograft

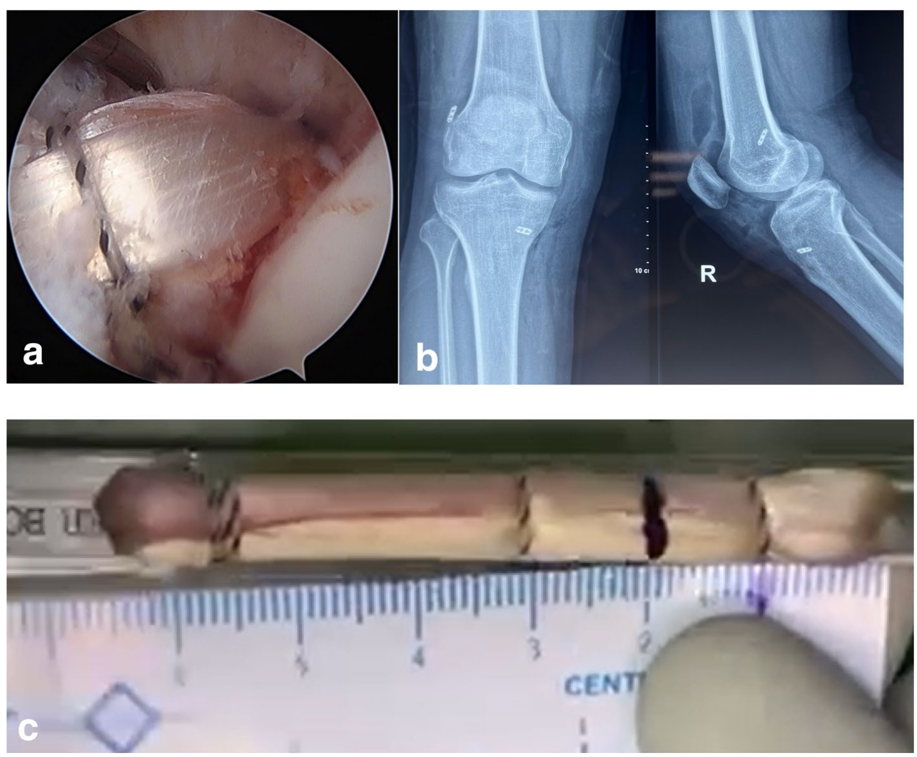

Fig. 3 Illustrating ACL reconstruction with HT autograft; **a** Intraoperative picture, **b** Postoperative X-ray, and **c** Prepared HT autograft

on the same day except in cases where meniscus repair was performed. Weight-bearing was delayed for 8 weeks in meniscus repair cases. The goal of rehabilitation was to achieve full range of motion by the end of 6 weeks and to achieve full athletic activity at the end of 9 months.

Statistical analysis

Sample size estimation was based on the Lysholm score (SD: 6–9) with an effect size of 5.3 at 1 year follow-up [9]. With an alpha error of 5% and a beta error of 20%, the study power was 80%, requiring 32 patients per group. Statistical analyses were performed using SPSS 22.0 (IBM Corp). Data normality was assessed with the Kolmogorov–Smirnov test. Between-group comparisons were conducted using the independent sample *t*-test for normally distributed continuous variables and the Mann–Whitney U test for non-normally distributed or ordinal data. Within-group pre-and postoperative comparisons were analyzed using the paired *t*-test or Wilcoxon signed-rank test, as appropriate. Nominal categorical data were assessed with the Pearson chi-square test. A *p*-value < 0.05 was considered statistically significant.

Result

A total of 69 patients were included in the final analysis (QT: 33, HT: 36). The demographic characteristics are presented in Table 1. Although patients in the QT group were younger, the difference was not statistically significant. Road traffic injury, primarily due to falls from two-wheelers, was the most common cause of ACL injury (44.9%), followed by sports-related trauma (34.7%). The mean graft diameter was significantly larger in the QT group compared to the HT group (9.22 ± 0.62 mm vs. 8.69 ± 0.63 mm, $p < 0.001$). In the HT group, graft diameter showed a significant negative correlation with Lachman grading (Spearman correlation coefficient, r = − 0.33, $p = 0.04$), whereas no significant correlation was observed in the QT group (r = 0.23, $p = 0.21$).

No significant difference was observed between the QT and HT groups in terms of clinical parameters and PROMs preoperatively. Preoperative clinical parameters and PROMs in both groups are shown in Table 2.

At the final follow-up, a negative Lachman test was observed in 72.7% of patients in the QT group compared to 41.6% in the HT group ($p = 0.04$). Similarly, a higher proportion of patients in the QT group demonstrated a negative anterior drawer test (81.8% vs. 47.2%, $p = 0.005$).

Table 1 Illustration of demographic profile of the patients

Parameters		QT (n = 33)	HT (n = 36)	*p* value
Age (years, mean ± SD)	29.42 ± 8.79	28.73 ± 8.80	30.06 ± 8.87	0.53
Sex	Male/female	30/3	32/4	0.78
Duration of symptoms (weeks)	10.68 ± 2.05	14.13 ± 1.97	7.52 ± 2.09	0.18
Mode of injury	RTI (fall from two-wheeler)	18	13	0.19
	RTI (auto rickshaw toppled down)	0	1	
	RTI (pedestrian hit by two-wheeler)	0	1	
	Sports related injury	9	15	
	Fall from height	3	5	
	Fall of heavy object over knee	3	0	
	Fall on uneven ground	0	1	

Table 2 Preoperative clinical parameters and PROMs

Graft type	QT				HT				*p* value
Grade	0	1	2	3	0	1	2	3	
Lachman*	0	1	24	8	0	4	28	4	0.07
Anterior drawer test*	0	6	18	9	0	5	27	4	0.39
Pivot shift test*[1]	0	17	12	1	6	15	9	2	0.33
VAS score**	1.58 ± 0.79				1.92 ± 1.27				0.19
Lysholm score**	53.36 ± 6.86				49.56 ± 15.35				0.21
KOOS score**	46.97 ± 9.21				50.03 ± 10.36				0.20

[1]Pivot shift could not be done in 3 cases in HT group and 4 cases in QT group

*Statistical analysis was performed using the Mann–Whitney U test

**Statistical analysis was performed using the independent sample t-test

Table 3 Clinical parameters at 6 and 12 months follow-up in both QT & HT group

Test	Follow-up	Graft type	Grade 0	Grade 1	Grade 2	Grade 3	*p*-value
Lachman test*	6 months	QT	24	6	3	0	0.06
		HT	18	12	6	0	
	12 months	QT	24 (72.7%)	6	3	0	0.04
		HT	15 (41.6%)	12	8	1	
Anterior drawer test*	6 months	QT	19	17	0	0	0.01
		HT	27	6	0	0	
	12 months	QT	27 (81.8%)	5	1	0	0.02
		HT	17 (47.2%)	15	4	0	
Pivot shift test*	6 months	QT	0	0	0	0	0.09
		HT	3	0	0	0	
	12 months	QT	33	0	0	0	0.27
		HT	31 (86.1%)	4	1	0	
Side-to-side difference in anterior translation** (mm)	12 months	QT	2.73 ± 1.64				0.23
		HT	3.77 ± 2.01				

(Grade 0 = Negative)

*Statistical analysis was performed using the Mann–Whitney U test

** Statistical analysis was performed using the independent sample t-test

Table 4 PROMs at 6 and 12 months follow-up in both QT & HT group

	6 months follow-up			12 months follow-up		
Graft type	QT	HT	*p* value	QT	HT	*p* value
VAS score*	0.18 ± 0.392	0.19 ± 0.525	0.91	0.07 ± 0.03	0.12 ± 0.04	0.49
Lysholm Score*	77.18 ± 7.243	71.28 ± 8.147	0.002	92.42 ± 3.36	88.15 ± 4.32	0.02
KOOS score*	74.91 ± 8.457	76.14 ± 5.530	0.47	94.13 ± 4.62	89.62 ± 4.28	0.03

*Statistical analysis was performed using the independent sample t-test.

Table 5 Preoperative versus postoperative PROMs in QT & HT group

Graft type	QT			HT		
	Preoperative	12 months follow-up	*p* value	Preoperative	12 months follow-up	*p* value
Lysholm Score*	53.55 ± 7.203	92.42 ± 3.36	< 0.001	49.56 ± 15.35	88.15 ± 4.32	< 0.001
KOOS score*	46.97 ± 9.36	94.13 ± 4.62	< 0.001	50.03 ± 10.36	89.62 ± 4.28	< 0.001

*Statistical analysis was performed using the paired t-test.

No significant difference was noted between the groups for the pivot shift test. Although the side-to-side anterior translation was greater in the HT group, the difference was not statistically significant (Table 3).

At 12 months, the QT group demonstrated significantly higher Lysholm and KOOS scores compared to the HT group (Table 4). No significant difference was observed in VAS scores between groups at any follow-up. Both groups showed significant improvement in Lysholm and KOOS scores from baseline to final follow-up (Table 5, Fig. 4).

Graft failure and donor site morbidity

Graft failure occurred in one patient from the HT group at 9 months postoperatively, confirmed on MRI, while no failures were observed in the QT group. Sensory hypoesthesia over the anteroinferior knee and anterolateral proximal leg was reported in 11 patients (30.5%) from the HT group, with no such deficits in the QT group. One patient in the QT group developed postoperative stiffness requiring arthroscopic arthrolysis.

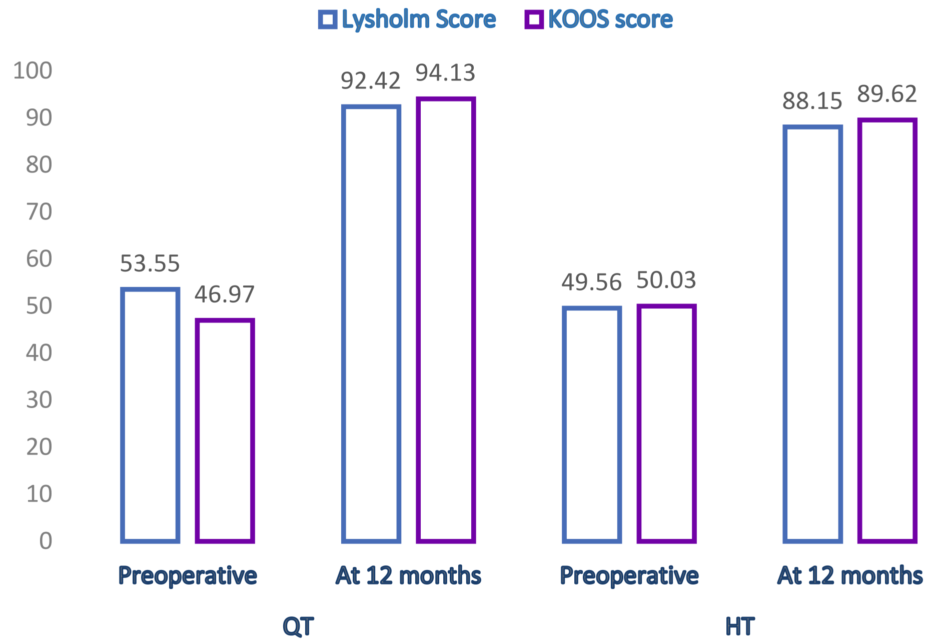

Fig. 4 Bar chart showing PROMs in preoperative versus at 12 months in QT and HT group

Discussion

At final follow-up, QT showed more negative Lachman (72.7% vs. 41.6%, $p = 0.049$) and anterior drawer tests (81.8% vs. 47.2%, $p = 0.005$), while HT showed a higher trend of positive pivot shift tests without statistical significance.

Clinical assessment of laxity

In our study, a higher proportion of patients in the QT group demonstrated negative Lachman and anterior drawer tests compared to the HT group, consistent with the findings of Cavaignac et al. [10], who reported negative Lachman tests in 93.1% of QT and 46.2% of HT cases ($p < 0.005$). Similarly, Karikis et al. [11] observed negative Lachman results in 50% of patients in the single-bundle HT group. In contrast, Lee et al. [12] reported no significant differences, with negative Lachman tests in 66.7% of QT and 70.8% of HT cases, and negative anterior drawer tests in 70.8% and 77.0%, respectively ($p > 0.05$). These findings differ from ours, likely due to their retrospective design and the use of double-bundle ACL reconstruction in the HT group.

At 12 months, 13.9% of HT patients demonstrated a positive pivot shift (Grade 1: 4, Grade 2: 1), while none were observed in the QT group; however, the difference was not statistically significant. Similarly, Cavaignac et al. [10] reported more negative pivot shift tests with QT (93.6% vs. 64.1%, $p = 0.052$), and Lind et al. [13] found comparable rates between QT (81%) and HT (77%) at 1 year ($p = 0.34$). These findings align with ours, indicating that QT autografts may offer superior restoration of rotational stability. As residual pivot shift is linked to poorer functional outcomes and increased risk of secondary meniscal or chondral injury, the absence of positive pivot shift in the QT group underscores its potential clinical advantage. The lower incidence of pivot shift in our study may relate to the larger diameter and cross-sectional area of QT grafts compared with HT. This is consistent with Lund et al. [14], who showed that the broader cross-sectional area of QT grafts more effectively fills the intra-articular space than flat BPTB grafts, thereby reducing the risk of a positive pivot shift.

The HT group demonstrated greater side-to-side anterior translation, though not statistically significant, consistent with Karpinski et al. [15] (1.56 vs. 1.64 mm, $p = 0.69$)) and Runer et al. [16] (1.9 vs. 2.1 mm, $p = 0.60$). In a prospective cohort study, Lubis et al. [17] reported greater laxity in the HT group than the QT group (3.87 mm vs. 3.12 mm, $p = 0.015$) using rollimeter measurement. On the contrary, Sofu et al. [18], in a retrospective study in 2013 (QT: 23, HT: 21), reported greater laxity with QT (5.56 vs. 3.67 mm, $p < 0.001$). This discrepancy may be due to advances in QT harvesting techniques since then and the lack of functional outcome measures in their study, limiting direct comparison.

Patient reported outcome measures (PROMs)

At 12 months, the QT group showed significantly higher Lysholm and KOOS scores compared with the HT group, although the QT group also started with higher preoperative Lysholm scores, and both groups demonstrated similar postoperative gains. Prior studies have reported comparable trends. Cavaignac et al. [10] noted significantly higher Lysholm (89 ± 6.9 vs 83.1 ± 5.3) and KOOS Sport (82 ± 11.3

vs 67 ± 12.4) scores with QT, while Brinkman et al. [19] found superior Lysholm scores at 5 years (92.6 vs. 88.0, $p = 0.007$). In contrast, Mouarbes et al. [20] and Karpinski et al. [15] observed only nonsignificant differences, though QT outcomes remained consistently higher. Taken together, these results support a tendency toward improved functional outcomes with QT autografts. However, in our study, the differences did not exceed the minimal clinically important difference (MCID) thresholds, suggesting that the statistical differences may hold limited clinical significance.

In our study, no significant differences in VAS scores were observed between the QT and HT groups at any follow-up. This aligns with the findings of Runer et al. [14], who reported comparable long-term pain outcomes, and Vilchez-Cavazos et al. [21], who similarly found no difference between graft types in a randomized controlled trial.

Graft size is an important factor in ACL reconstruction. In our study, the QT graft had a significantly larger diameter than the HT graft (9.22 vs. 8.69 mm, $p < 0.001$), consistent with Brinkman et al. [19], who also reported larger diameters with QT. The use of the all-inside technique likely contributed to the comparatively greater diameter observed in both graft types in our study. Also, larger graft diameters were associated with lower Lachman grades, indicating better anterior stability. This finding is consistent with previous reports, where smaller hamstring grafts (< 8 mm) were linked to increased residual laxity and graft failure, while larger grafts correlated with improved outcomes on Lachman and pivot shift testing [22, 23].

Graft failure and donor site morbidity

In our study, the only case of graft failure occurred in the HT group. This finding is in line with Runer et al. [24], who also reported a single graft failure in the HT group over 2 years of follow-up and later concluded that QT grafts are associated with a lower rupture rate compared with HT grafts [25]. Sensory hypoesthesia in the distribution of the infrapatellar branch of the saphenous nerve was observed in 30.5% of patients in the HT group, consistent with reported rates of 21–83% in hamstring harvest using the anteromedial approach [26], while no such cases were noted in the QT group. One patient in the QT group developed knee stiffness requiring arthroscopic arthrolysis, a complication also reported by Brinkman et al. [19], who observed arthrofibrosis in 8.1% of QT and 6.5% of HT reconstructions without significant difference.

The QT graft may offer certain advantages over the HT graft in ACL reconstruction, particularly in terms of donor site morbidity and knee stability. These potential differences could be explained by their distinct anatomical and biomechanical characteristics, with the HT composed of parallel fibers and the QT consisting of multiple layers [27–29]. Nonetheless, further research with larger patient cohorts, especially among high-risk groups such as young individuals and elite athletes, is needed to better define revision rates and long-term outcomes.

The current study has limitations. First, the study's major limitation is its short follow-up period. Longer follow-up could establish the incidence of graft failure and other long-term complications associated with QT grafts. Second, the lack of patient and assessor blinding may have introduced performance and detection bias in subjective outcomes and stability grading, affecting the reliability of group comparisons. Third, the assessment of quadriceps strength and knee range of motion was not conducted, hence constraining the comprehensiveness of the functional outcome comparison between graft types. Finally, as a single-center study, the findings may have limited external validity.

Conclusion

Quadriceps tendon autografts in arthroscopic ACL reconstruction provided superior translational stability and lower donor site morbidity compared with hamstring autografts. While PROM differences were below the MCID, QT grafts showed a trend toward better rotational stability and may be preferable in patients concerned about sensory deficits. Longer-term studies are required to confirm the risks of stiffness and graft failure.

Author contributions Arvind Kumar yadav and Md Quamar Azam made conceptualization and design of the study. Data curation was performed by Ajay Sharma and Nirvin Paul. Arvind Kumar yadav and Anurag Bhakhar drafted the manuscript. Manuscript editing and review was done by Bhaskar Sarkar. All authors read and approved the final manuscript.

Funding The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

Data availability The data that support the findings of this study are available on request from the corresponding author (XXBLINDEDXX).

Declarations

Conflict of Interest The authors declare no conflict of interests.

Ethics approval The study has been approved by Institutional Ethics Committee (XXBLINDEDXX/IEC/23/240).

Informed consent Informed and written consent was obtained from each participant before enrolment in the study.

Permission to reproduce material from other sources Not applicable.

Clinical trial registration The study has been registered under CTRI with registration number: CTRI/2023/07/055894.

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